

# **EXP8:WIRESHARK SEQ ANALYSIS**

**RAVI BALA M.V**

**Register Number: 24BCE1419**

**Faculty Name: Subbulakshmi**

**Lab Slot: L55 + L56**

**Course Name: Computer Networks Lab**

**Course Code: BCSE308L**

## RELATIVE SEQUENCE NUMBER IS ZERO FOR (SYN)

```
▼ Transmission Control Protocol, Src Port: 51497, Dst Port: 443, Seq: 0, Len: 0
  Source Port: 51497
  Destination Port: 443
  [Stream index: 14]
  [Stream Packet Number: 1]
  ▶ [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 0]
  Sequence Number: 0 (relative sequence number)
  Sequence Number (raw): 2607725924
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 0
  Acknowledgment number (raw): 0
  1000 .... = Header Length: 32 bytes (8)
  ▶ Flags: 0x002 (SYN)
  Window: 65535
  [Calculated window size: 65535]
  Checksum: 0x2b82 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  ▶ Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP), SACK permitted
  ▶ [Timestamps]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
```

## RELATIVE SEQ=0 AND ACK=1(SYN-ACK)

```
4 Frame 642: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface \Device\NPF_{6600A9D0-F702-4A6A-BA40-647C98BB210E}, id 0
  Ethernet II, Src: Intel_06:4e:74 (dc:45:46:06:4e:74), Dst: Intel_06:4e:74 (dc:45:46:06:4e:74)
  Internet Protocol Version 6, Src: 2001:500:88:200:10, Dst: 2401:4900:4df2:f787:21e2:8755:b195:f681
  ▼ Transmission Control Protocol, Src Port: 443, Dst Port: 53826, Seq: 0, Ack: 1, Len: 0
    Source Port: 443
    Destination Port: 53826
    [Stream index: 15]
    [Stream Packet Number: 2]
    ▶ [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 0]
    Sequence Number: 0 (relative sequence number)
    Sequence Number (raw): 3133077345
    [Next Sequence Number: 1 (relative sequence number)]
    Acknowledgment Number: 1 (relative ack number)
    Acknowledgment number (raw): 1472649340
    1000 .... = Header Length: 32 bytes (8)
    ▶ Flags: 0x012 (SYN, ACK)
    Window: 22080
    [Calculated window size: 22080]
    Checksum: 0x3853 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    ▶ Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, SACK permitted, End of Option List (EOL), End of Option List (EOL)
    ▶ [Timestamps]
    ▶ [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
```

## RELATIVE SEQ=1 AND ACK=1(ACK)

```
4 Frame 639: Packet, 1454 bytes on wire (11632 bits), 1454 bytes captured (11632 bits) on interface \Device\NPF_{6600A9D0-F702-4A6A-BA40-647C98BB210E}, id 0
  Ethernet II, Src: Intel_06:4e:74 (dc:45:46:06:4e:74), Dst: b2:52:e9:42:44:96 (b2:52:e9:42:44:96)
  Internet Protocol Version 6, Src: 2401:4900:4df2:f787:21e2:8755:b195:f681, Dst: 2001:500:88:200:10
  ▼ Transmission Control Protocol, Src Port: 51497, Dst Port: 443, Seq: 1, Ack: 1, Len: 1380
    Source Port: 51497
    Destination Port: 443
    [Stream index: 14]
    [Stream Packet Number: 4]
    ▶ [Conversation completeness: Complete, WITH_DATA (31)]
    [TCP Segment Len: 1380]
    Sequence Number: 1 (relative sequence number)
    Sequence Number (raw): 2607725925
    [Next Sequence Number: 1381 (relative sequence number)]
    Acknowledgment Number: 1 (relative ack number)
    Acknowledgment number (raw): 3856202954
    0101 .... = Header Length: 20 bytes (5)
    ▶ Flags: 0x010 (ACK)
    Window: 255
    [Calculated window size: 65280]
    [Window size scaling factor: 256]
    Checksum: 0x01da [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    ▶ [Timestamps]
    ▶ [SEQ/ACK analysis]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
```

## DEMONSTRATION OF ATTACKS:

|      |           |                      |                     |          |      |      |                                     |
|------|-----------|----------------------|---------------------|----------|------|------|-------------------------------------|
| 2016 | 73.014047 | 10.77.167.47         | 35.228.141.16       | TCP      | 66   |      | 60533 → 443 [SYN] Seq=0             |
| 2017 | 73.225803 | 35.228.141.16        | 10.77.167.47        | TCP      | 70   | 2016 | 0.211756000 443 → 60533 [SYN, ACK]  |
| 2018 | 73.225943 | 10.77.167.47         | 35.228.141.16       | TCP      | 54   | 2017 | 0.000140000 60533 → 443 [ACK] Seq=1 |
| 2019 | 73.226574 | 10.77.167.47         | 35.228.141.16       | TCP      | 1454 |      | 60533 → 443 [ACK] Seq=1             |
| 2020 | 73.226574 | 10.77.167.47         | 35.228.141.16       | TLSv1... | 446  |      | Client Hello (SNI=e2c13             |
| 2021 | 73.233325 | 2603:1040:a06:3::... | 2401:4900:4df2:f... | TLSv1... | 105  |      | Application Data                    |

THE [SYN,ACK] GREY COLOR SHOWS THE You will see many "SYN" packets without any corresponding "SYN-ACK" or "ACK."

|      |           |                     |                     |          |     |      |                                                                     |
|------|-----------|---------------------|---------------------|----------|-----|------|---------------------------------------------------------------------|
| 2117 | 78.510357 | 2001:4860:4827:f... | 2401:4900:4df2:f... | UDP      | 92  |      | 443 → 54266 Len=26                                                  |
| 2118 | 78.691025 | 35.228.141.16       | 10.77.167.47        | TLSv1... | 97  |      | Application Data                                                    |
| 2119 | 78.691306 | 10.77.167.47        | 35.228.141.16       | TCP      | 54  | 2118 | 0.000281000 60533 → 443 [FIN, ACK] Seq=2559 Ack=4959 Win=65280 Len= |
| 2120 | 78.911743 | 35.228.141.16       | 10.77.167.47        | TLSv1... | 82  | 2119 | 0.220437000 Application Data                                        |
| 2121 | 78.911743 | 35.228.141.16       | 10.77.167.47        | TCP      | 58  |      | 443 → 60533 [FIN, ACK] Seq=4983 Ack=2560 Win=61952 Len=             |
| 2122 | 78.911855 | 10.77.167.47        | 35.228.141.16       | TCP      | 54  | 2120 | 0.000112000 60533 → 443 [RST, ACK] Seq=2560 Ack=4983 Win=0 Len=0    |
| 2123 | 79.620102 | 10.77.167.47        | 10.77.167.180       | DNS      | 77  |      | Standard query 0x9b57 AAAA beacons3.gvt2.com                        |
| 2124 | 79.638987 | 10.77.167.180       | 10.77.167.47        | DNS      | 77  |      | Standard query response 0x9b57 AAAA beacons3.gvt2.com               |
| 2125 | 80.172171 | 2401:4900:4df2:f... | 2404:6800:4007:8... | QUIC     | 801 |      | Protected Payload (KP0), DCID=fd0c7efed1d74aa6                      |

Packets with the [RST] flag set in red.Sends Reset (R) flags to force a connection to close.

|      |            |                      |                      |          |     |      |                                                               |
|------|------------|----------------------|----------------------|----------|-----|------|---------------------------------------------------------------|
| 3035 | 140.558816 | 2401:4900:40f2:f...  | 2404:6800:4007:8...  | UDP      | 91  |      | 55214 → 443 Len=29                                            |
| 3036 | 140.459011 | 2404:6800:4007:8...  | 2401:4900:4df2:f...  | UDP      | 87  |      | 443 → 53214 Len=25                                            |
| 3037 | 140.940239 | 2603:1061:f:100::... | 2401:4900:4df2:f...  | TCP      | 78  |      | 443 → 49321 [RST, ACK] Seq=5745 Ack=3393 Win=0 Len=0          |
| 3038 | 140.980731 | 2401:4900:4df2:f...  | 2603:1040:a06:3::... | TLSv1... | 105 |      | Application Data                                              |
| 3039 | 141.062104 | 2603:1040:a06:3::... | 2401:4900:4df2:f...  | TCP      | 78  | 3038 | 0.081373000 443 → 51780 [ACK] Seq=244 Ack=451 Win=536 Len=0   |
| 3040 | 141.087355 | 2603:1061:e::254     | 2401:4900:4df2:f...  | TCP      | 78  |      | 443 → 50895 [RST, ACK] Seq=5747 Ack=4295 Win=0 Len=0          |
| 3041 | 141.172701 | 2401:4900:4df2:f...  | 2001:4860:4827:7...  | UDP      | 859 |      | 54266 → 443 Len=797                                           |
| 3042 | 141.262949 | 2001:4860:4827:7...  | 2401:4900:4df2:f...  | UDP      | 99  |      | 443 → 54266 Len=33                                            |
| 3043 | 141.275350 | 2401:4900:4df2:f...  | 2001:4860:4827:7...  | UDP      | 95  |      | 54266 → 443 Len=33                                            |
| 3044 | 141.306895 | 2001:4860:4827:7...  | 2401:4900:4df2:f...  | UDP      | 757 |      | 443 → 54266 Len=691                                           |
| 3045 | 141.308664 | 2001:4860:4827:7...  | 2401:4900:4df2:f...  | UDP      | 110 |      | 443 → 54266 Len=44                                            |
| 3046 | 141.321481 | 2401:4900:4df2:f...  | 2001:4860:4827:7...  | UDP      | 102 |      | 54266 → 443 Len=40                                            |
| 3047 | 141.364228 | 34.1.15.89           | 10.77.167.47         | TCP      | 58  |      | [TCP Keep-Alive] 443 → 55795 [ACK] Seq=4805 Ack=1889 Wi       |
| 3048 | 141.364329 | 10.77.167.47         | 34.1.15.89           | TCP      | 54  |      | [TCP Keep-Alive ACK] 55795 → 443 [ACK] Seq=1889 Ack=480       |
| 3049 | 141.382703 | 2001:4860:4827:7...  | 2401:4900:4df2:f...  | UDP      | 92  |      | 443 → 54266 Len=26                                            |
| 3050 | 143.094354 | 2401:4900:4df2:f...  | 2603:1046:c04:89...  | TCP      | 74  |      | 65376 → 443 [RST, ACK] Seq=2391 Ack=6377 Win=0 Len=0          |
| 3051 | 143.212967 | 2404:6800:4007:8...  | 2401:4900:4df2:f...  | UDP      | 103 |      | 443 → 65022 Len=37                                            |
| 3052 | 143.234058 | 2401:4900:4df2:f...  | 2404:6800:4007:8...  | UDP      | 95  |      | 65022 → 443 Len=33                                            |
| 3053 | 145.195373 | 2404:6800:4007:8...  | 2401:4900:4df2:f...  | UDP      | 219 |      | 443 → 53214 Len=157                                           |
| 3054 | 145.218582 | 2401:4900:4df2:f...  | 2404:6800:4007:8...  | UDP      | 95  |      | 53214 → 443 Len=33                                            |
| 3055 | 145.273659 | 2404:6800:4007:8...  | 2401:4900:4df2:f...  | UDP      | 219 |      | 443 → 53214 Len=157                                           |
| 3056 | 145.274199 | 2401:4900:4df2:f...  | 2404:6800:4007:8...  | UDP      | 96  |      | 53214 → 443 Len=34                                            |
| 3057 | 147.203753 | 10.77.167.47         | 103.10.124.5         | TLSv1... | 108 |      | Application Data                                              |
| 3058 | 147.326727 | 103.10.124.5         | 10.77.167.47         | TCP      | 54  | 3057 | 0.122974000 443 → 63919 [ACK] Seq=2497 Ack=498 Win=1869 Len=0 |
| 3059 | 148.268310 | 204.79.197.222       | 10.77.167.47         | TCP      | 58  |      | 443 → 49667 [RST, ACK] Seq=5191 Ack=4656 Win=0 Len=0          |
| 3060 | 151.134360 | 2401:4900:4df2:f...  | 2603:1063:14::128    | TLSv1... | 124 |      | Application Data                                              |
| 3061 | 151.288559 | 2603:1040:a06:3::... | 2401:4900:4df2:f...  | TLSv1... | 105 |      | Application Data                                              |
| 3062 | 151.319283 | 2603:1063:14::128    | 2401:4900:4df2:f...  | TLSv1... | 113 | 3060 | 0.184923000 Application Data                                  |

You send a SYN to a "Reflector" server, but you spoof the source address as the "Victim." The Reflector sends the SYN-ACK to the Victim, not you.Incoming SYN-ACKs to a machine that never sent a SYN.